

$f_{t+k} = g(f_t, f_{t+1}, \dots)$

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$$f_H = g(f_t, \{t-1, \dots\})$$

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- The diagram illustrates a recurrent neural network structure. It shows three input boxes labeled $\text{input}(t-2)$, $\text{input}(t-1)$, and $\text{input}(t)$. Arrows from $\text{input}(t-2)$ and $\text{input}(t-1)$ point to a box labeled hidden . An arrow from the hidden box points to the $\text{input}(t)$ box.

$$t \rightarrow \boxed{f(t)} \rightarrow$$

Example: Recurrent networks

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- The diagram illustrates the architecture of a recurrent neural network (RNN) in two parts. The left part shows a single time step t where an input X_t is fed into an RNN block, which also receives information from a 'Memory' block at time $t-1$. The RNN block produces an output Y_t and updates the 'Memory' block for time t . The right part shows a sequence of three time steps, with inputs x_t, x_{t+1}, x_{t+2} and hidden states h_t, h_{t+1}, h_{t+2} connected sequentially. The word 'store' is written below the first hidden state, and a red arrow at the top indicates the progression of time.

- Distributed hidden state: can efficiently 'past' information
- Nonlinear dynamics: model complex relationships
- Temporal and accumulative: can build sequential semantics